

California Housing Price Prediction

Machine Learning Project · Regression with Random Forest

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This project builds a machine learning model that predicts the median house value of a California house from census data such as income, location and housing characteristics. It covers the full workflow: data inspection, feature engineering, preprocessing, model training and evaluation, all packaged into a single reusable scikit-learn pipeline. An interactive Power BI dashboard then lets anyone explore the same data by location and income on a bubble map or heat map.

0.81

R² score

\$32.3K

Mean absolute error

\$50.0K

Root mean squared error

Project at a Glance

Goal	Predict the median house value (in USD) of a California house
Problem type	Supervised learning, regression
Dataset	California Housing: 20,640 houses, 10 columns, 1 target (median_house_value)
Model	Random Forest Regressor (scikit-learn)
Tools	Python, pandas, NumPy, scikit-learn, Matplotlib, Jupyter Notebook, Power BI
Evaluation	R ² , MAE and RMSE on a 20% hold-out test set (4,128 houses)
Dashboard	Interactive Power BI dashboard with bubble and heat map views, two slicers and KPI cards

About the Data

The dataset describes 20,640 California houses using census-style information. Each row is a house, not an individual house.

- **Location:** longitude, latitude and ocean_proximity (the only text column).
- **Housing:** housing_median_age, total_rooms and total_bedrooms.
- **People:** population, households and median_income.
- **Target:** median_house_value, the value the model learns to predict.

Data quality was good overall. Only total_bedrooms had missing values (207 of 20,640 rows), which are filled with the median during preprocessing.

Approach: Start to End

01 Load & inspect	02 Engineer features	03 Stratified split	04 Preprocess	05 Random Forest	06 Evaluate
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Stage	What was done	Why it matters
1. Load & inspect	Loaded housing.csv and checked its shape, data types, missing values and first rows.	Confirms data quality before modelling. Found 207 missing values in total_bedrooms (about 1%).
2. Feature engineering	Created rooms_per_household, bedrooms_per_room and population_per_household.	Raw totals depend on district size. Ratios describe how people actually live and compare better across districts.
3. Income categories	Grouped median_income into 5 bands (0–2, 2–4, 4–6, 6–8, 8+).	Used only to guide the split, since income is a key driver of house prices.
4. Exploration	Plotted histograms of every numeric column.	Shows how values are spread and highlights skew and unusual ranges.
5. Stratified split	80/20 train/test split with StratifiedShuffleSplit on the income bands (random_state = 42).	The test set mirrors the income mix of the full data, so evaluation is fair and reproducible.
6. Preprocessing	Numeric columns: standard scaling and median imputation. Text column (ocean_proximity): one-hot encoding. Combined with ColumnTransformer.	Handles missing values and text data, and applies identical transformations to training and test data.
7. Model pipeline	Chained the preprocessor and a RandomForestRegressor into one scikit-learn Pipeline.	One object to train and predict with. It also prevents data leakage from the test set.
8. Train & predict	Fitted the pipeline on 16,512 training rows and predicted 4,128 unseen test rows.	Measures how the model performs on data it has never seen.
9. Evaluation	Computed R ² , MAE and RMSE, then plotted actual vs predicted prices against a perfect-prediction line.	Gives both numeric scores and a visual check of prediction quality.

Results

Metric	Score	What it tells us
R ² score	0.8137	The model explains about 81% of the variation in house prices on unseen data.
MAE	\$32,331.02	On average, a prediction is off by roughly \$32K.
RMSE	\$49,961.68	Higher than MAE because it penalises large errors more. It shows that a few districts are predicted with bigger misses.

The final plot compares actual and predicted prices against a perfect-prediction line, making it easy to see where the model is accurate and where it drifts.

Interactive Dashboard (Power BI)

After building the model, I created an interactive dashboard called California House Data to explore the same dataset visually. It lets anyone filter the districts by location and income level and immediately see how prices, income and housing figures change on a map, in either a bubble view or a heat map view.

What the dashboard contains

Component	What it does
Interactive map	Every bubble is one district, placed by its longitude and latitude on a Bing map. You can zoom from the whole state down to a single city, as in Figure 2.
Heat map view	The same map can be switched to a heat map. Instead of individual bubbles, it shades the map by how densely districts cluster, so hotspots stay visible where bubbles overlap (Figure 3).
Location Category slicer	Button filter based on ocean_proximity: <1H OCEAN, INLAND, ISLAND, NEAR BAY and NEAR OCEAN. Click one or more to focus on a type of location.
Income_Category slicer	Buttons 1 to 5 for the income bands created in the notebook (1 is the lowest, 5 is median income above 8). It is the same income_cat feature used for the stratified split.
KPI cards	Ten cards summarise the districts currently shown. Averages are used for house value, income, the three ratios and housing age. Sums are used for population, households, total rooms and total bedrooms.
Cross-filtering	Slicers, map and cards are linked. Any filter or map selection updates everything at once.

Reading the unfiltered view

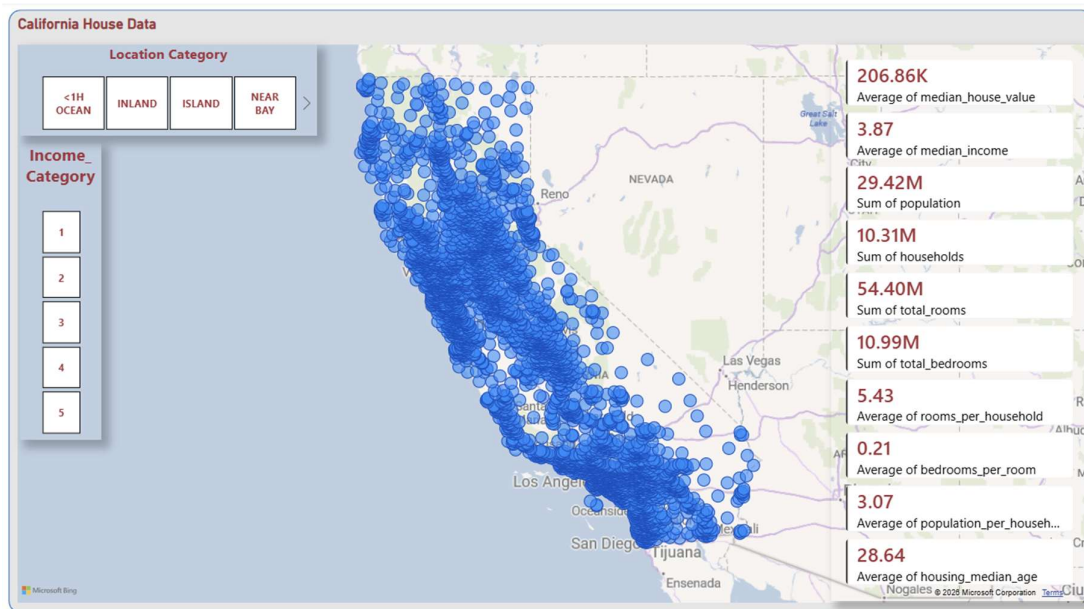


Figure 1. Dashboard with no filters applied, showing every district in California.

- **Map:** the bubbles trace the Pacific coast and the Central Valley, and are densest around the Bay Area, Greater Los Angeles and San Diego. The eastern deserts and the Sierra Nevada are almost empty, which shows how strongly people and housing are concentrated near the coast.
- **Price and income:** the average district has a median house value of 206.86K and a median income of 3.87.

- **Housing:** a typical district has 5.43 rooms and 3.07 people per household, and its median housing age is 28.64 years.
- **Totals:** about 29.42M people in 10.31M households, with 54.40M rooms and 10.99M bedrooms.

Reading a filtered view

In Figure 2, the slicers are set to NEAR BAY and Income_Category 5, the highest income band, and the map is zoomed into the San Francisco Bay Area.

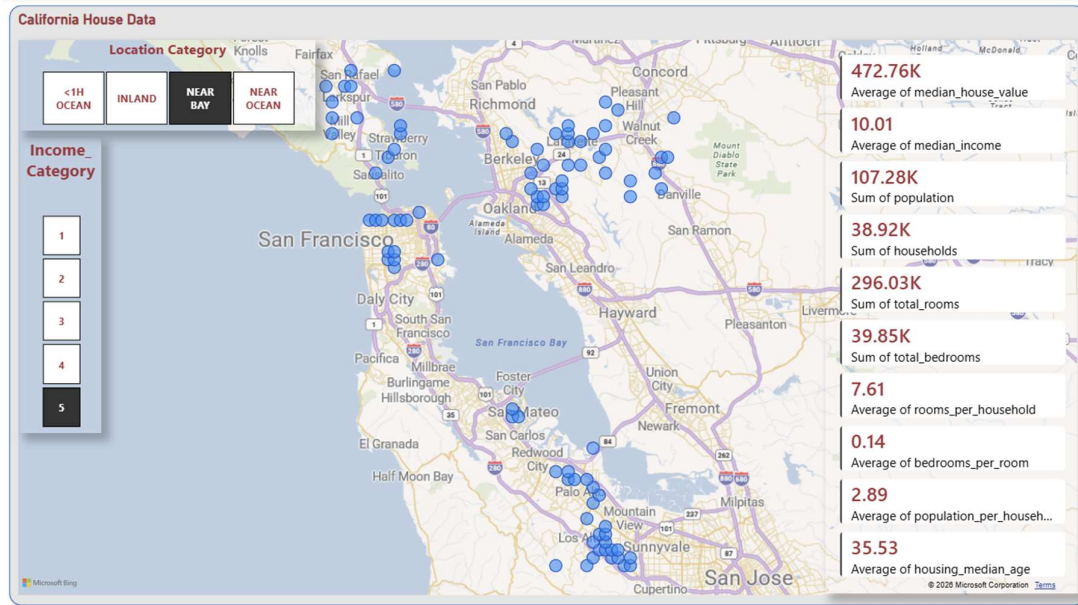


Figure 2. Dashboard filtered to NEAR BAY districts in the highest income band (5), bubble view.

- **Map:** the points sit in Marin County, San Francisco, the East Bay around Oakland, Berkeley and Walnut Creek, and the Peninsula down to Palo Alto and Sunnyvale.
- **Price and income:** these districts are far more expensive. The average house value is 472.76K, about 2.3 times the statewide figure, and the average income is 10.01 against 3.87. This average is also close to the roughly \$500K ceiling in the dataset's house values, which is why capped data is listed as a limitation.
- **Housing:** homes are larger (7.61 rooms per household against 5.43), have fewer bedrooms per room (0.14 against 0.21), hold slightly fewer people (2.89 against 3.07) and are older (35.53 years against 28.64).
- **Scale:** the combined population is 107.28K, only about 0.4% of the statewide total.

Measure	Figure 1: no filters	Figures 2 and 3: Near Bay, income 5
Average median_house_value	206.86K	472.76K
Average median_income	3.87	10.01
Average rooms_per_household	5.43	7.61
Average bedrooms_per_room	0.21	0.14
Average population_per_household	3.07	2.89
Average housing_median_age	28.64	35.53
Sum of population	29.42M	107.28K
Sum of households	10.31M	38.92K

Heat map view

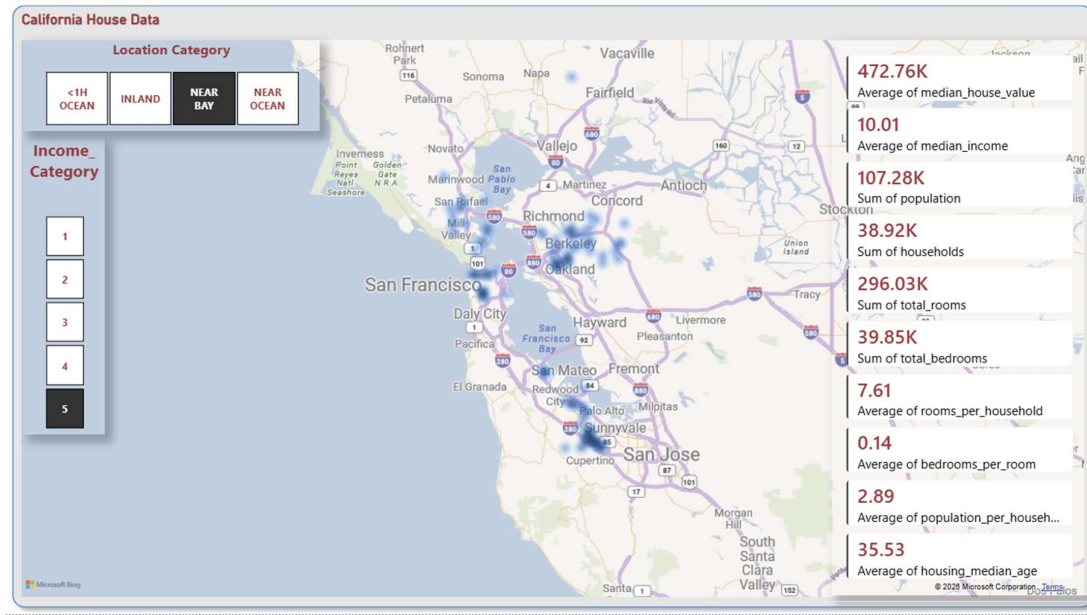


Figure 3. The same filtered selection shown as a heat map.

- **Hotspots:** the darkest patches are around Sunnyvale and Cupertino in the south, the Oakland and Berkeley area, and San Francisco. These are where the high-income Near Bay districts are packed most tightly.
- **Why it helps:** in the bubble view, overlapping points hide how many districts sit in the same place. The heat map blends them into one intensity scale, so crowded areas are easier to read. The bubble view is better for spotting individual districts.
- **Consistency:** the slicers and cards are unchanged between Figures 2 and 3, so switching map view does not change the numbers.

Together, the views show why location and income matter for price prediction. Districts near the Bay with the highest incomes sit at more than double the statewide average price, and these are the same factors the model uses, through latitude, longitude, ocean_proximity and median_income.

How it was built

- **Data:** the dashboard uses the same California housing data, including the income category and the three engineered ratio columns created in the notebook.
- **Map:** a map visual plots each district using its longitude and latitude, with a heat map view of the same points.
- **Slicers:** two button slicers, one for ocean_proximity and one for income_cat, control every visual on the page.
- **Cards:** ten KPI cards summarise whatever is selected. I set each one to the right calculation, with averages for rate-type fields and sums for totals, so every figure is meaningful.
- **Styling:** a consistent maroon and blue-grey theme, shaded panels and a clear title keep the page easy to read.

Recommendations

The dashboard already works well as an exploration tool. But as I am in beginner phase I would do these changes after getting my hands comfortable to make it more accurate and more useful:

Recommendation for myself	Why it helps
Use friendlier card titles and formats	Some titles are cut off, such as Average of population_per_househ.... Short names (for example, Avg people per household) and a currency format for house value would read more clearly.
Add a house count card	Averages and totals change with every filter. A count of houses shows how many records each view is based on, since a selection like Near Bay with income 5 is far smaller than the whole state.
Show price on the map	The bubbles and the heat map show where houses are, not how expensive they are. Colour or size the bubbles by median_house_value, and weight the heat map by price if the visual allows it.
Add supporting charts	A bar chart of average price by location category, a scatter of income against house value and a price histogram would back up the map with clear comparisons.
Bring in the model's predictions	Export the test-set predictions and show actual against predicted value, or error size, on the map. This connects the dashboard to the machine learning work and shows where the model struggles.
Add tooltips and a reset button	Hover tooltips with house details and a one-click Clear filters button make exploring easier for first-time viewers.
Tidy the slicer layout	Only four of the five location options are visible at once, so one is always hidden behind the scroll arrow. Resizing the slicer to show all five avoids this.
Publish and share	Publish to the Power BI Service and add the link to your portfolio, so recruiters can try the dashboard themselves.

Skills Demonstrated

- **Data analysis:** inspecting data quality, handling missing values and visual exploration.
- **Feature engineering:** turning raw counts into meaningful ratios.
- **Sound ML practice:** stratified sampling, a held-out test set and a leakage-free pipeline.
- **Model evaluation:** choosing and interpreting R^2 , MAE and RMSE.
- **Data visualisation:** building an interactive Power BI dashboard with a bubble map, heat map, slicers and KPI cards.
- **Clean, reproducible code:** fixed random seeds and a single end-to-end scikit-learn pipeline.

Limitations

- **Single split:** scores come from one StratifiedShuffleSplit, so they may shift with a different split. Cross-validation would give a steadier estimate.
- **No tuning or comparison:** the Random Forest uses default settings and was not benchmarked against other models.
- **Aggregated data:** predictions are for districts, so the model cannot price an individual property.

- **Dated and capped data:** this public dataset is based on 1990 census figures and caps house values at about \$500K, so it is not suited to today's market or very expensive areas.
- **Limited interpretability:** a Random Forest is hard to explain, and no feature-importance analysis was done yet.

Future Scope

Improvement	Expected benefit
Cross-validation and hyperparameter tuning	More reliable scores, and better accuracy using GridSearchCV or RandomizedSearchCV.
Benchmark other models	Compare against Linear/Ridge Regression and Gradient Boosting, XGBoost or LightGBM to justify the final choice.
Dashboard upgrade	Apply the dashboard recommendations and show model predictions and errors on the map.
Explainability	Use feature importance or SHAP to show which factors drive predictions.
Richer features	Add distance to coast or major cities and cluster latitude/longitude to capture location better.
Deployment	Save the pipeline with joblib and serve it through a Streamlit app or a Flask/FastAPI endpoint.
Reproducibility	Add a requirements file, unit tests and experiment tracking so anyone can rerun the work.
Newer data	Retrain on recent housing data so predictions reflect today's market.

Conclusion

This project shows a complete and clean machine learning workflow, from raw data to a tested model that explains about 81% of price variation on unseen data. The Power BI dashboard makes the same data easy to explore for non-technical viewers. Together they form a solid baseline, and the improvements above would make it more accurate, explainable and ready for real-world use.

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